

Q1.

$2 - 3i$ is one root of the equation

$$z^3 + mz + 52 = 0$$

where m is real.

(a) Find the other roots.

(3)

(b) Determine the value of m .

(2)

(Total 5 marks)

Q2.

$$p(z) = z^4 + 3z^2 + az + b, \quad a \in \mathbb{R}, b \in \mathbb{R}$$

$2 - 3i$ is a root of the equation $p(z) = 0$

(a) Express $p(z)$ as a product of quadratic factors with real coefficients.

(5)

(b) Solve the equation $p(z) = 0$.

(1)

(Total 6 marks)

Q3.

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(a) Solve the equation $w^2 + 6w + 34 = 0$, giving your answers in the form $p + qi$, where p and q are integers.

(3)

(b) It is given that $z = i(1 + i)(2 + i)$.

(i) Express z in the form $a + bi$, where a and b are integers.

(3)

(ii) Find integers m and n such that $z + mz^* = ni$.

(3)

(Total 9 marks)

Q4.

(a) It is given that $z = x + yi$, where x and y are real numbers.

(i) Write down, in terms of x and y , an expression for $(z - 2i)^*$.

(1)

(ii) Solve the equation

$$(z - 2i)^* = 4iz + 3$$

giving your answer in the form $a + bi$.

(5)

- (b) It is given that $p + qi$, where p and q are real numbers, is a root of the equation $z^2 + 10iz - 29 = 0$.

Without finding the values of p and q , **state** why $p - qi$ is **not** a root of the equation $z^2 + 10iz - 29 = 0$.

(1)

(Total 7 marks)

Q5.

- (a) Solve the following equations, giving each root in the form $a + bi$:

(i) $x^2 + 9 = 0$;

(1)

(ii) $(x + 2)^2 + 9 = 0$.

(1)

- (b) (i) Expand $(1 + x)^3$.

(1)

- (ii) Express $(1 + 2i)^3$ in the form $a + bi$.

(3)

- (iii) Given that $z = 1 + 2i$, find the value of

$$z^* - z^3$$

(2)

(Total 8 marks)

Q6.

- (a) Express in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$:

(i) $4(1 + i\sqrt{3})$;

(ii) $4(1 - i\sqrt{3})$.

(3)

- (b) The complex number z satisfies the equation

$$(z^3 - 4)^2 = -48$$

Show that $z^3 = 4 \pm 4\sqrt{3}i$.

(2)

- (c) (i) Solve the equation

$$(z^3 - 4)^2 = -48$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(5)

- (ii) Illustrate the roots on an Argand diagram.

(3)

- (d) (i) Explain why the sum of the roots of the equation

$$(z^3 - 4)^2 = -48$$

is zero.

(1)

- (ii) Deduce that $\cos \frac{\pi}{9} + \cos \frac{3\pi}{9} + \cos \frac{5\pi}{9} + \cos \frac{7\pi}{9} = \frac{1}{2}$.

(3)

(Total 17 marks)

Q7.

- (a) (i) It is given that α and β are the roots of the equation

$$x^2 - 2x + 4 = 0$$

Without solving this equation, show that α^3 and β^3 are the roots of the equation

$$x^2 + 16x + 64 = 0$$

(6)

- (ii) State, giving a reason, whether the roots of the equation

$$x^2 + 16x + 64 = 0$$

are real and equal, real and distinct, or non-real.

(2)

- (b) Solve the equation

$$x^2 - 2x + 4 = 0$$

(2)

- (c) Use your answers to parts (a) and (b) to show that

$$(1 + i\sqrt{3})^3 = (1 - i\sqrt{3})^3$$

(2)

(Total 12 marks)

Q8.

- (a) Solve the following equations, giving each root in the form $a + bi$:

- (i) $x^2 + 16 = 0$;

(2)

(ii) $x^2 - 2x + 17 = 0$.

(2)

(b) (i) Expand $(1 + x)^3$

(2)

(ii) Express $(1 + i)^3$ in the form $a + bi$.

(2)

(iii) Hence, or otherwise, verify that $x = 1 + i$ satisfies the equation

$$x^3 + 2x - 4i = 0$$

(2)

(Total 10 marks)

Q9.

The cubic equation

$$z^3 + pz^2 + 6z + q = 0$$

has roots α , β , and γ .

(a) Write down the value of $\alpha\beta + \beta\gamma + \gamma\alpha$.

(1)

(b) Given that p and q are real and that $\alpha^2 + \beta^2 + \gamma^2 = -12$:

(i) explain why the cubic equation has two non-real roots and one real root;

(2)

(ii) find the value of p .

(4)

(c) One root of the cubic equation is $-1 + 3i$.

Find:

(i) the other two roots;

(3)

(ii) the value of q .

(2)

(Total 12 marks)

Q10.

(a) (i) Given that $z^6 - 4z^3 + 8 = 0$, show that $z^3 = 2 \pm 2i$.

(2)

(ii) Hence solve the equation

$$z^6 - 4z^3 + 8 = 0$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(6)

- (b) Show that, for any real values of k and θ ,

$$(z - ke^{i\theta})(z - ke^{-i\theta}) = z^2 - 2kz \cos \theta + k^2$$

(2)

- (c) Express $z^6 - 4z^3 + 8$ as the product of three quadratic factors with real coefficients.

(3)

(Total 13 marks)

Q11.

The cubic equation

$$x^3 + px^2 + qx + r = 0$$

where p , q and r are real, has roots α , β and γ .

- (a) Given that

$$\alpha + \beta + \gamma = 4 \quad \text{and} \quad \alpha^2 + \beta^2 + \gamma^2 = 20$$

find the values of p and q .

(5)

- (b) Given further that one root is $3 + i$, find the value of r .

(5)

(Total 10 marks)

Q12.

It is given that $z = e^{i\theta}$.

- (a) (i) Show that

$$z + \frac{1}{z} = 2 \cos \theta$$

(2)

- (ii) Find a similar expression for

$$z^2 + \frac{1}{z^2}$$

(2)

- (iii) Hence show that

$$z^2 - z + 2 - \frac{1}{z} + \frac{1}{z^2} = 4 \cos^2 \theta - 2 \cos \theta$$

(3)

- (b) Hence solve the quartic equation

$$z^4 - z^3 + 2z^2 - z + 1 = 0$$

giving the roots in the form $a + ib$.

(5)

(Total 12 marks)

Q13.

The function f is defined by

$$f(x) = \frac{x^2 + 2x + 2}{x^2}$$

- (a) Write down the equations of the two asymptotes to the curve $y = f(x)$.

(2)

- (b) By considering the expression $x^2 + 2x + 2$:

- (i) show that the graph of $y = f(x)$ does not intersect the x -axis;

(2)

- (ii) find the non-real roots of the equation $f(x) = 0$.

(3)

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- (c) (i) Show that, if the equation $f(x) = k$ has two equal roots, then

$$4 - 8(1 - k) = 0$$

(3)

- (ii) Deduce that the graph of $y = f(x)$ has exactly one stationary point and find its coordinates.

(4)

(Total 14 marks)